

1.

What is the main use of Resource Allocation Graph (RAG) ?

- A. To allocate memory**
- B. To manage CPU time**
- C. To detect and avoid deadlocks**
- D. To prevent paging**

Answer: C

2.

In deadlock prevention, how is the hold and wait condition prevented

- A. Allowing processes to request all resources at once**
- B. Allowing circular wait**
- C. Not allowing mutual exclusion**
- D. Terminating all processes**

Answer: A

3.

Which of the following IPC mechanisms allows communication between processes on different machines?

- A. Pipe**
- B. Socket**
- C. Shared Memory Model**
- D. Message Queue**

Answer: B

4.

Which of the following is a drawback of deadlock prevention?

- A. Requires too much memory**
- B. It makes the system non-deterministic**
- C. Leads to low resource utilization**
- D. Causes compiler error**

Answer: C

5.

The major difference between semaphore and mutex is:

- A. Mutex can be shared between processes**
- B. Only one thread can acquire a mutex at a time, semaphores can allow, multiple**
- C. Semaphores are not used in synchronization**
- D. Mutex is used in message passing**

Answer: B

6.

In semaphores, if a process calls wait() and the semaphore value is 0, what happens?

- A. Process executes normally**
- B. Process is blocked until the value is positive**
- C. Process crashes**
- D. Process enters the critical section**

Answer: B

7.

If $S = 5$, and four wait(s) operations followed by two signal(s) operations executed, what is the final value of S ?

- A. 1
- B. 2
- C. 3
- D. 5

Answer: C